Customer Shopping Behavior Analysis

1. Project Overview

Understanding how customers shop is one of the most valuable capabilities a business can develop. This report presents the findings of a detailed analysis of customer shopping behavior, drawing on transactional data from 3,900 individual purchases spread across multiple product categories.

The primary objective of this project was to move beyond surface-level sales figures and instead explore the underlying patterns driving consumer decisions. Specifically, the analysis set out to answer four key questions:

- What do customers spend, and how does that vary by age, gender, or geography?
- Which customer segments are most valuable, and how can they be better retained?
- How do pricing strategies such as discounts and promotions influence purchasing behavior?
- What product categories and individual items generate the most satisfaction and revenue?

The insights generated from this analysis are intended to support data-driven decisions in marketing, product strategy, and customer retention. The project followed a structured workflow — starting with raw data cleaning in Python, followed by business-level querying in SQL, and culminating in a visual dashboard built in Power BI.

2. Dataset Overview

The dataset used in this analysis contains 3,900 rows and 18 columns, capturing a wide range of information about each transaction. Below is an overview of the key variable groups:

2.1 Variable Categories

- Customer Demographics — Age, Gender, Location, and Subscription Status
- Purchase Details — Item Purchased, Category, Purchase Amount (USD), Season, Size, and Color
- Shopping Behavior — Discount Applied, Promo Code Used, Previous Purchases, Purchase Frequency, Review Rating, and Shipping Type

2.2 Data Quality

The dataset was largely complete, with only one notable data quality issue: 37 missing values in the Review Rating column. These were later imputed using the median rating per product category, ensuring that the missingness did not skew product performance metrics. No other columns contained null values.

2.3 Key Statistics at a Glance

Metric	Value
Total Transactions	3,900
Average Purchase Amount	$59.76
Age Range	18 to 70 years
Average Customer Age	44 years
Average Review Rating	3.75 / 5.00
Missing Values	37 (Review Rating only)

3. Data Preparation and Cleaning

Before any analysis could begin, the raw dataset required a thorough cleaning and transformation process. This was carried out in Python using the pandas library. The steps taken are described below.

3.1 Initial Exploration

The dataset was loaded into a pandas DataFrame and inspected using df.info() and .describe(). This initial review confirmed the data types, flagged the missing values in the Review Rating column, and provided a statistical overview of all numeric fields. The purchase amount ranged from $20 to $100, with a median of $60, suggesting a relatively consistent price distribution without extreme outliers.

3.2 Handling Missing Values

Rather than removing records with missing review ratings — which would have meant losing nearly 1% of the dataset — a category-level imputation strategy was used. Each missing rating was replaced with the median rating of the corresponding product category. This approach preserves data integrity while avoiding the distortion that a global median imputation might introduce.

3.3 Column Standardization and Feature Engineering

All column names were renamed to snake_case format to ensure consistency across Python and SQL environments. Two new features were also created to enrich the dataset:

- age_group — Customer ages were binned into four segments: Young Adult (18–30), Adult (31–45), Middle-aged (46–60), and Senior (61+). This made age-based analysis significantly cleaner and more interpretable.
- purchase_frequency_days — Derived from the frequency of purchases field, this column expressed how often each customer made a purchase in terms of days, enabling more precise behavioral segmentation.

3.4 Redundancy Check

A consistency check was performed on the discount_applied and promo_code_used columns, which were found to be functionally identical — every customer who used a promo code also had a discount applied, and vice versa. To avoid multicollinearity and simplify downstream analysis, the promo_code_used column was dropped.

3.5 Database Integration

Once the cleaned DataFrame was finalized, it was loaded directly into a PostgreSQL database using SQLAlchemy. This allowed for structured SQL querying in the next phase of the project without any manual data transfer.

4. Business Analysis Using SQL

With the cleaned data stored in PostgreSQL, a series of targeted queries were run to answer specific business questions. The ten analyses are presented below, each followed by a summary of findings.

4.1 Revenue by Gender

A simple but telling comparison — male customers generated more than double the revenue of female customers over the same period.

Gender	Total Revenue (USD)
Female	$75,191
Male	$157,890

This gap likely reflects a difference in transaction volume rather than spend per order, given that the average purchase amount across genders is similar. The business may want to investigate whether marketing efforts are reaching female customers as effectively, or whether product mix plays a role.

4.2 High-Spending Discount Users

A common concern with discount programs is that they primarily attract low-margin customers. This analysis disproved that assumption. Of the total customer base, 839 customers applied a discount and still spent above the overall average purchase amount. This suggests that discounts, in many cases, are being used by customers who would have spent significantly regardless — meaning discounts may be acting as an unnecessary cost rather than a conversion driver for this segment.

4.3 Top 5 Products by Customer Rating

Customers were asked to rate their purchases on a scale of 1 to 5. The five products that received the highest average ratings were:

Rank	Product	Average Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82

Rank	Product	Average Rating
4	Hat	3.80
5	Skirt	3.78

It is worth noting that even the top-rated products fall just below 4.0 out of 5.0. While ratings are acceptable, there is meaningful room for improvement. Qualitative feedback collection — through post-purchase surveys — could help identify specific pain points that are suppressing ratings across the board.

4.4 Shipping Type Comparison

Customers who chose Express shipping spent slightly more on average than those who selected Standard shipping — a difference of roughly $2 per transaction.

Shipping Type	Average Purchase Amount
Standard	$58.46
Express	$60.48

This is a modest but consistent pattern. Express shipping customers may reflect a higher-intent buyer segment — people who are more committed to their purchase and willing to pay for faster delivery. Targeting promotions or loyalty perks toward this group could yield strong returns.

4.5 Subscribers vs. Non-Subscribers

The subscription analysis revealed a somewhat counterintuitive result: non-subscribers not only outnumber subscribers by nearly three to one, but they also spend very slightly more per transaction.

Subscription Status	Total Customers	Avg. Spend	Total Revenue
Subscriber	1,053	$59.49	$62,645
Non-Subscriber	2,847	$59.87	$170,436

The near-identical average spend per group suggests that subscribers are not currently receiving enough exclusive value to differentiate their experience. The business should consider strengthening the subscription value proposition — through early access, exclusive discounts, or personalized recommendations — to convert more of the 2,847 non-subscribers.

4.6 Discount-Dependent Products

Identifying which products rely most heavily on discounts to drive purchases is critical for margin management. The five products with the highest discount attachment rates were:

Rank	Product	Discount Rate (%)
1	Hat	50.00%
2	Sneakers	49.66%
3	Coat	49.07%
4	Sweater	48.17%
5	Pants	47.37%

These products see discounts applied in roughly half of all purchases, raising questions about whether their listed prices are competitive or whether customers have been conditioned to wait for a sale. A pricing strategy review for these items — potentially including small permanent price reductions — could reduce discount dependency without sacrificing demand.

4.7 Customer Segmentation by Purchase History

Customers were classified into three segments based on their purchase history: New (fewer than 5 previous purchases), Returning (5 to 10), and Loyal (more than 10). The breakdown was:

Segment	Number of Customers
Loyal	3,116
Returning	701
New	83

The dominance of the Loyal segment is a positive sign — the business has successfully retained a large portion of its customer base over time. However, the very small number of New customers (83) suggests that acquisition may be a growing concern. Balancing investment between retention and new customer growth will be important going forward.

4.8 Top 3 Products per Category

Within each of the four main product categories, the three most frequently purchased items were identified. This provides a clear view of what customers are actually buying, which should inform both inventory planning and promotional focus.

Category	Rank 1	Rank 2	Rank 3
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	—

Jewelry, Blouses, and Jackets lead their respective categories. These items are natural candidates for featuring prominently in campaigns and ensuring they remain consistently in stock.

4.9 Repeat Buyers and Subscription Behavior

Of the 3,476 customers who made more than five purchases, only 958 hold an active subscription — while 2,518 do not. This is a significant opportunity. Frequent buyers are already demonstrating strong brand loyalty through their purchase behavior, yet the majority have not converted to a subscription. A targeted re-engagement campaign offering a free trial or introductory discount could efficiently grow the subscriber base without needing to attract new customers.

4.10 Revenue by Age Group

Revenue contribution by age group was relatively balanced, though Young Adults led the ranking:

Age Group	Total Revenue (USD)
Young Adult (18–30)	$62,143
Middle-Aged (46–60)	$59,197
Adult (31–45)	$55,978
Senior (61+)	$55,763

Young Adults contributing the most revenue is worth noting, especially since they are also the most likely segment to respond to digital marketing, loyalty apps, and social commerce. Targeted campaigns for this group — particularly around seasonal trends — could strengthen revenue further.

5. Power BI Dashboard

To make the findings accessible to non-technical stakeholders, an interactive dashboard was developed in Microsoft Power BI. The dashboard provides a single, consolidated view of the most important metrics and supports filtering by Subscription Status, Gender, Product Category, and Shipping Type.

Key metrics displayed on the dashboard include the total number of customers (3,900), the average purchase amount ($59.76), and the average review rating (3.75). Supporting visuals break down revenue and sales volume by category and age group, and a donut chart shows the 73%/27% split between non-subscribers and subscribers at a glance.

The interactive filters make it easy for team members in marketing, operations, or product to slice the data according to their specific area of focus — without requiring any SQL or Python knowledge.

Figure 1: Customer Behavior Dashboard (Power BI)

6. Business Recommendations

Based on the analysis, five actionable recommendations have been identified for the business:

6.1 Convert Repeat Buyers into Subscribers

With over 2,500 high-frequency buyers currently outside the subscription program, this is the single largest opportunity identified in the analysis. A dedicated re-engagement campaign — offering a trial period or first-month discount — targeted specifically at customers with more than five purchases could meaningfully grow the subscriber base with relatively low acquisition cost.

6.2 Redesign the Subscription Value Proposition

Subscribers and non-subscribers spend almost the same amount per transaction, which suggests the current subscription benefits are not compelling enough to change behavior. The business should consider what exclusive, high-perceived-value benefits can be added — such as priority customer service, free express shipping, or early access to new collections.

6.3 Revisit Pricing for Discount-Heavy Products

Hats, Sneakers, Coats, Sweaters, and Pants all see discounts applied in nearly half of all purchases. This level of discount dependency erodes margins and may signal that base prices are set too high relative to what customers perceive as fair value. A strategic review — potentially including modest permanent price reductions alongside stricter discount controls — is advisable.

6.4 Invest in Customer Acquisition

With only 83 new customers in the dataset, the business appears to be heavily dependent on its existing loyal customer base. While retention is valuable, a healthy business needs a steady pipeline of new customers to replace natural churn. Increased investment in acquisition channels — particularly digital campaigns targeting Young Adults — would help balance this.

6.5 Leverage Top-Rated and Best-Selling Products

Products like Gloves, Sandals, Jewelry, and Jackets consistently perform well on both satisfaction and sales volume. These items deserve prominent placement in marketing materials, email campaigns, and the website homepage. Highlighting customer ratings alongside these products can also serve as social proof to drive further conversion.

7. Conclusion

This analysis has painted a detailed and actionable picture of how customers interact with this business. The data reveals a fundamentally healthy customer base — one that is largely loyal, reasonably satisfied, and consistent in its spending. At the same time, several areas of untapped potential have been surfaced: subscription conversion among repeat buyers, discount dependency in key product lines, and a thin pipeline of new customer acquisition.

The recommendations outlined in this report are grounded directly in the data and are prioritized to deliver the greatest impact with the most efficient use of resources. Implementing even two or three of these changes — particularly around subscription growth and discount policy — could have a meaningful effect on revenue and profitability within a single quarter.

As a next step, the team recommends presenting these findings to the relevant department heads and establishing clear ownership for each recommendation, along with measurable success criteria to track progress over the months ahead.

8. SQL Queries

The following screenshots capture all ten SQL queries executed in PostgreSQL, along with the Python code used for exploratory data analysis and data cleaning. Each query was written to answer a specific business question, as described throughout this report.

Query 1: Revenue by Gender

```sql
--Q1 What is the total revenue generated by male vs. female customers?
select gender, sum(customer.purchase_amount) as Total_Revenue
from customer
group by gender
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page N

	gender text	total_revenue numeric
1	Female	75191
2	Male	157890

Query 2: High-Spending Discount Users

```sql
--Q2 Which customers ised a discount but still spent more than
select customer_id, purchase_amount
from customer
where discount_applied = 'Yes' and purchase_amount > (select avg(purchase_amount) from customer)
```

Data Output Messages Notifications

Showing rows: 1 to 839 Page No: 1 of 1

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62

Query 3: Top 5 Products by Rating

```sql
--Q3. Which are the top 5 products with the highest average review rating?
select item_purchased, ROUND(avg(review_rating::numeric),2) as "Average Product Rating"

from customer
group by item_purchased
order by avg(review_rating) desc
limit 5;
```

Data Output Messages Notifications

SQL Showing rows: 1 to 5 Page No: 1 of

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Query 4: Shipping Type Comparison

```sql
--Q4. Compare the average Purchase Amounts between Standard and Express Shipping.
select shipping_type,
ROUND(avg(purchase_amount),2)
from customer
where shipping_type in ('Standard', 'Express')
group by shipping_type

```

Data Output Messages Notifications

SQL Showing rows: 1 to 2 Page No: 1

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Query 5: Subscribers vs. Non-Subscribers

```sql
--Q5. Do subscribed customers spend more?
--Compare average spend and total revenue between subscribers and non-subscribers.
select subscription_status,
count(customer_id) as total_customer,
Round(avg(purchase_amount),2) as avg_spend,
Round(Sum(purchase_amount),2) as total_revenue
from customer
group by subscription_status
order by total_revenue, avg_spend desc
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1

	subscription_status text	total_customer bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Query 6: Discount-Dependent Products

```sql
--Q6. Which 5 products have the highest percentage of purchases with discounts applied?
select item_purchased,
Round(100 * sum(case when discount_applied = 'Yes' then 1 else 0 end)/count(*),2) as discount_rate
from customer
group by item_purchased
order by discount_rate desc
limit 5;
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

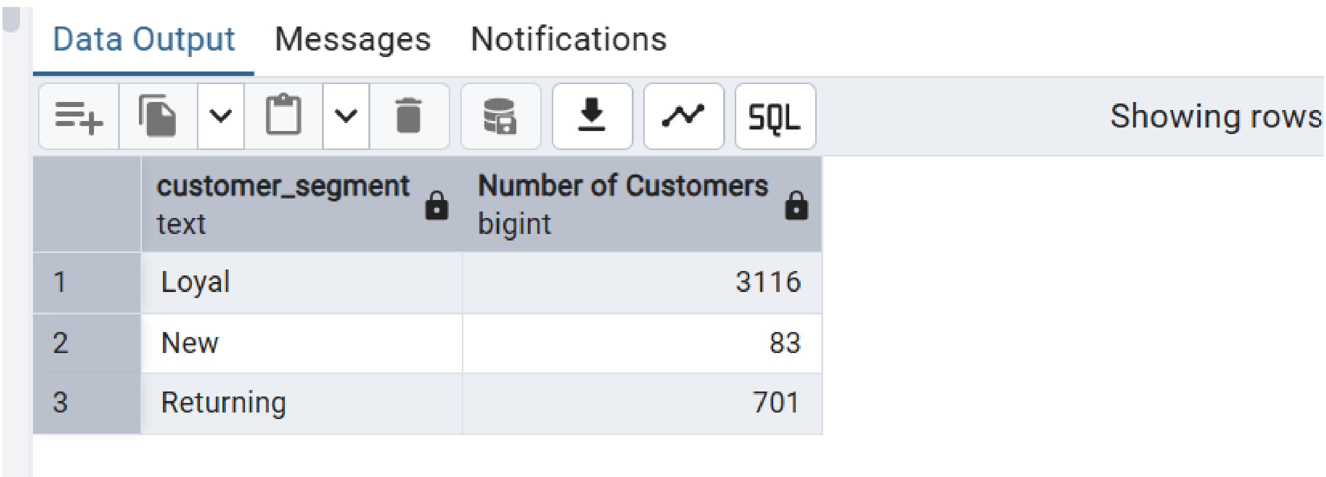
	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

Query 7: Customer Segmentation

```sql
--Q7. Segment customers into New, Returning, and Loyal based on their
--total number of previous purchases, and show the count of each segment.

with customer_type as(
select customer_id,
previous_purchases,
case when previous_purchases = 1 then 'New'
     when previous_purchases between 2 and 10 then 'Returning'
     else 'Loyal'
     end as customer_segment
from customer
)
select customer_segment, count(*) as "Number of Customers"
from customer_type
group by customer_segment
```

Query 8: Top 3 Products per Category



	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Query 8: Top 3 Products per Category

```sql
--Q8. What are the top 3 most purchased products within each category?

with item_counts as(
select category,
item_purchased,
count(customer_id) as total_orders,
ROW_NUMBER() over(partition by category order by count(customer_id) desc) as item_rank
from customer
group by category , item_purchased
)
select item_rank, category, item_purchased, total_orders
from item_counts
where item_rank <= 3;
```

Data Output Messages Notifications

Showing rows: 1 to 11

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Query 9: Repeat Buyers and Subscriptions

```sql
--Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?
select subscription_status,
count(customer_id) as repeate_buyers
from customer
where previous_purchases >5
group by subscription_status
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	subscription_status text	repeate_buyers bigint
1	No	2518
2	Yes	958

Query 10: Revenue by Age Group

```sql
select age_group, sum(purchase_amount) as revenue
from customer
group by age_group
order by revenue desc
```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No

	age_group text	revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Appendix: Tools and Technologies Used

Tool / Technology	Purpose
Python (pandas, SQLAlchemy)	Data cleaning, feature engineering, and database loading
PostgreSQL	Structured SQL querying for business analysis
Power BI	Interactive dashboard and data visualization
SQL (queries)	Revenue analysis, segmentation, product ranking